K-Nearest Neighbours (KNN) Regression Assingment

Answers

The given dataset,

Operating Temperature (°C)	Fuel Consumption (L/hr)
40	12
45	14
50	15
55	18
60	20
65	22
70	25
75	27

Now to find ~~KNN~~ the predicted fuel consumption when operating temperature is T = 58°C, we need to find the distance between the given temperature and that dataset temperature. then the given dataset changes to

Operating Temperature(°c)	Fuel Consumption (L/hr)	Distance = |58° - Operating temp|
40	12	|58-40| = 18
45	14	|58-45| = 13
50	15	|58-50| = 8
55	18	|58-55| = 3 ✓
60	20	|58-60| = 2 ✓
65	22	|58-65| = 7 ✓
70	25	|58-70| = 12
75	27	|58-75| = 17

From analysing the new table, we realise that,

When operating temperature is 60°c the distance is 2 and hence fuel consumption is 20.

When operating temperature is 55°c the distance is 3 and hence fuel consumption is 18.

When operating temperature is 65°c the distance is 7 and hence fuel consumption is 22.

We are taking 3 least value because it is given k = 3 in the question

Now, k = 3

$$\text{Prediction} = \frac{20 + 18 + 22}{3} \quad \text{(Average)}$$

$$= \frac{60}{3}$$

$$= 20 \text{ L/hr}$$

∴ For operating temperature 58°c, the fuel consumption is 20 L/hr.